Yuankun Huang

Software Engineer, Game Systems & Simulation

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Unity/C# client engineer who builds the systems other engineers ship on. Five years in live-service mobile games: core client engineer on Mythic Heroes from early development to 10M+ downloads, and live-event framework lead on Top Museum. Next focus: real-time simulation, grounded in an offline physics bake built at work and deterministic replay in C++20.

CORE SKILLS

Languages	C# (production since 2020), C++20 (personal projects), .NET 8
Unity & Client	Unity (live-service mobile), Timeline/Animator, Addressables, ILRuntime hot-update, Play Mode regression tests, Unity Profiler, Frame Debugger, editor tooling
Simulation	Collision detection, offline batch simulation, deterministic replay, multi-agent simulation, spatial partitioning
Networking & Build	Google Protobuf, SignalR, MessagePack, CMake, GitHub Actions CI, Cursor (team agentic workflow)

PROFESSIONAL EXPERIENCE

Unity Developer May 2025 – Present
IGG Canada, client development on two live-service mobile games *Vancouver, BC*

Top Museum, mobile strategy game (test release on Google Play) Feb 2026 – Present

- **Live-event framework.** Lead the C#/Unity client framework on which six other engineers have shipped 30+ event types; new events plug in through a content factory and registries, with no hub code changes.
- **Architecture standard.** Catalogued ten recurring anti-patterns in the event module and encoded a layering standard in the team's Cursor agent workflow, so AI-generated features stay within it.
- **13 event types.** Built them end to end, including guild duel, demolition, tribe and wolf, and extracted the participation and warm-up base four guild events share.
- **2D collision simulator.** Designed and built a double-precision ball simulator for a pachinko-style event: closed-form ballistic motion between contacts and swept collision detection on a 1/240 s sub-step.
- **Offline batch simulation.** Ran 5M seeded launches across CPU cores and kept only paths that met design rules, so designers set odds from measured outcomes.
- **Binary track replay.** Packed the 7,693 accepted paths into a 1.1 MB file; the server draws each outcome and the client replays a matching track, so every device shows the same path and runs no physics.
- **Regression testing.** Wrote Play Mode regression tests for 12 event types (78 cases) on the team's test framework, plus a nightly runner; they caught a bug that dropped rewards when a design change put two in the same second.
- **Showcase build size.** Automated stripping of hero showcase scenes (Timeline/Animator) on every regeneration; a frame-by-frame comparison rolls back any visible difference.
- **Tooling and mentoring.** Built a UI-layer prefab validator (editor tooling) for every UI generator. Mentor new engineers.

Mythic Heroes, live-service mobile RPG (10M+ downloads) Jun 2025 – Dec 2025

- **Pet progression.** Owned the pet line (summon, training, refinement, bonds), built the flip-card event and event calendar, and shipped 25 new UI windows through monthly ILRuntime hot-updates.

Career Break: Bereavement & Family Care *Feb 2025 – May 2025*

Software Engineer (Contract) Jan 2024 – Jan 2025
The HIVE, UBC Faculty of Medicine, gamified stroke rehabilitation *Vancouver, BC*

- **Sole engineer.** Replaced an early prototype of TrAIT, a stroke-rehabilitation game with Ultraleap hand tracking, with a ground-up rebuild; it succeeds the Kinect version validated in *Neurorehabilitation and Neural Repair*.
- **Accessible input.** Made the game playable with limited arm reach: multi-stage calibration maps each patient's reach to the full screen, and dwell-based selection replaces gestures.
- **Training rules.** Built 10-level streak progression and a restart-proof daily session cap rebuilt from on-device logs.
- **Research data.** Gave researchers per-frame movement data as CSV logs, zipped and emailed to the lab over SMTP.
- **Designer tooling.** Built a ScriptableObject config pipeline with editor code generation, so the team's designer tuned all 10 levels without code changes.

Unity Developer Sep 2020 – Sep 2024

IGG Canada, Mythic Heroes (live-service mobile RPG, 10M+ downloads) *Vancouver, BC*

- **Launch to live ops.** Core client engineer from early development through global launch and live ops.
- **Chat.** Owned the Chat client end to end: channels, direct messages, emoji, moderation and server-pushed presence over the game's Google Protobuf connection.
- **Guilds.** Co-owned the Guilds client, including client networking and multiplayer state sync over Protobuf.
- **UI framework.** Co-developed the shared UI framework behind the game's screens and authored 150+ of its windows.
- **Low-end Android.** Kept long lists and leaderboards smooth: profiled with the Unity Profiler and Frame Debugger, then added off-screen culling and caching and moved sprite atlases to Addressables.
- **Hot updates.** Shipped monthly live content through the ILRuntime hot-update pipeline.

SELECTED PROJECTS

Rhythm Fruit Shop [GitHub] *C++20 / SFML / CMake / GitHub Actions*

- **Deterministic replay.** Audio clock taken from the PCM playback cursor; GitHub Actions CI requires replay to reproduce the recorded result exactly and a perfect run to match a reference scoring implementation.
- **Allocation guard.** A custom operator-new guard asserts that the headless session core's input and update path never allocates; CMake dependency guards fail the build if a core layer links SFML or miniaudio.

Kodama [GitHub] *.NET 8 / SignalR / MessagePack / Unity*

- **Multi-agent simulation.** Server-authoritative in .NET 8: 10K agents on a 10 Hz tick in SoA component storage, with a behaviour pass measured at zero allocations.
- **Spatial partitioning.** Uniform grid for nearest-resource queries, checked against a linear scan on 60K randomized cases; state streams over SignalR and MessagePack to a GPU-instanced Unity client.

EDUCATION & CERTIFICATION

M.S. Computer Science (in progress, part-time), Georgia Institute of Technology 2026 – Present

Master of Digital Media, Centre for Digital Media, Vancouver 2021

B.Eng. Telecommunication Engineering, Queen Mary University of London (joint programme with BUPT) 2019

Unity Certified Expert: Programmer